Sumedh Ramakrishna



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Profile

I am a quantitative economist with a Master's in Economics from Heidelberg University, Germany. I combine formal training in econometric identification and statistical inference with hands-on experience training supervised models in Python, R, and SQL—including tree-based ensembles, regularized linear estimators, and neural sequence architectures. Currently deepening my foundation in large language model design and NLP, I am seeking roles where I can apply rigorous modeling to data science, financial analytics, or applied quantitative research.

I am **eligible to work** in the following countries without visa sponsorship: Germany, India, United Kingdom (2-year HPI visa), Netherlands (1-year Orientation visa).

Education

Heidelberg University *11/2025*
M.Sc. Economics (Grade: 2.0 - Good)

Shiv Nadar University *06/2021*
B.Sc. (Research) in Economics

Research & Analytical Experience

Master's Thesis (Grade: 1.8) *07/2025 – 11/2025*
Alfred Weber Institute (Prof. Stefan Klonner) Heidelberg, Germany

- Analyzed household investment behavior and financial commitment decisions using complex microdata panels from the Indonesian Family Life Survey (IFLS).
- Engineered a quantitative indexing method to structure multi-round longitudinal data, implementing non-linear econometric regressions and causal inference to evaluate household resource allocation under financial constraints.
- Executed reproducible data-cleaning and transformation pipelines in R and Stata across tens of thousands of survey records.

Master's Seminar (Decision Theory & Predictive Modeling) *04/2024 – 08/2024*
Alfred Weber Institute (Dr. Sebastian Ebert) Heidelberg, Germany

- Modeled choice behavior and risk-preference shifts across asymmetric risk environments, evaluating predictive accuracy of contextual choice models against standard expected-utility benchmarks.
- Structured decision trees and non-linear classification models to benchmark risk profiles and identify non-standard decision patterns relevant to credit risk assessment.

Research Intern *05/2021 – 11/2021*
Madras Inst. of Dev. Studies (Dr. L. Venkatachalam) Chennai, India

- Conducted statistical valuation and cost-benefit estimations for regional ecological assets to support structured policy and public investment allocations.
- Synthesized multi-source environmental and survey datasets into structured regression models and executive analytical reports.

Bachelor's Thesis *08/2019 – 05/2020*
Shiv Nadar University (Dr. Arka Roy Chaudhuri) Greater Noida, India

- Investigated macroeconomic liquidity shocks and informal cash circulation anomalies using large-scale National Sample Survey (NSS) expenditure microdata.
- Identified transient consumption spikes using difference-in-differences and time-series controls, evaluating systemic liquidity shifts.

Technical Skills

- **Programming & Tools:** Python, R, SQL, Stata, Git/GitHub, LaTeX.
- **Econometrics & Causal Inference:** Microeconometric estimation, panel data modeling, fixed effects, difference-in-differences, instrumental variables, hypothesis testing.
- **Machine Learning:** Supervised & unsupervised learning, mathematical optimization (gradient descent, loss formulation), regularization (L1/L2), tree ensembles (LightGBM, Random Forest), model evaluation & cross-validation.
- **Deep Learning & NLP:** Sequence modeling (LSTM/BiLSTM), dense embeddings, attention mechanisms, transformer architectures & LLM foundations.
- **Languages:** English (Native / Bilingual), German (B1, in progress), Tamil (Native / Bilingual).